Distributional Dynamics and Integration Order Governance: A Unified Cointegration Audit of U.S. Wage Share Measures and Capital Composition Interactions (1929–2021)

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July 2026

Abstract

This paper presents a forensic econometrics audit of the stochastic properties of U.S. wage share measures and their structural interactions with capital composition ($\tau_t = k_t^{\text{ME}} - k_t^{\text{NRC}}$) across nearly a century of macroeconomic accumulation (1929–2021). Grounded in classical-Marxian political economy and the UMass Amherst applied econometrics tradition, we evaluate four distinct wage share concepts: Non-Financial Corporate GVA Wage Share ($\omega_{\text{NFC,GVA}}$), Non-Financial Corporate NVA Wage Share ($\omega_{\text{NFC,NVA}}$), Total Corporate GVA Wage Share ($\omega_{\text{CORP,GVA}}$), and the unblocked Supervisory-Adjusted Shaikh-Tonak Wage Share ($\omega_{\text{ST,Tier2}}$).

Using the complete battery of unit root and stationarity tests from the R `urca` package—Augmented Dickey-Fuller (ADF), Elliott-Rothenberg-Stock DF-GLS, Phillips-Perron (PP), and KPSS—alongside Andrews-Zivot (AZ) endogenous structural break unit root tests, we establish three principal empirical findings. First, all wage share series are level-stationary or trend-stationary $I(0)$ processes around structural breaks, refuting the assumption that functional distribution follows an unconstrained random walk. Second, Andrews-Zivot endogenous break tests locate the primary structural breakpoint for the supervisory-adjusted Shaikh wage share ($\omega_{\text{ST,Tier2}}$) strictly at **1968** ($t_{\text{AZ}} = -5.151, p < 0.05$), aligning precisely with the late-Fordist profit squeeze zero-crossing identified in physical capital composition curvature ($\frac{d^2\hat{\tau}_t}{dt^2} = 0$). Third, we demonstrate that multiplying the $I(1)$ capital composition state variable (τ_t) by $I(0)$ wage share measures yields well-behaved $I(1)$ interaction terms ($\tau_t \times \omega_t \sim I(1)$), passing the Integration Order Gate for long-run cointegration estimation without incurring $I(2)$ explosion risks.

1 Introduction: Distributional Conflict and Integration Governance

In heterodox political economy, functional income distribution is not an exogenous policy parameter or a neutral residual. It is the core indicator of class struggle that conditions capital accumulation, capacity utilization, and choice of technique. When labor secures a higher share of net output (ω_t), the profit rate ($\pi_t = 1 - \omega_t$) falls, prompting capitalists to substitute machinery for labor to reduce unit labor costs and restore profitability.

To model this distribution-conditioned accumulation frontier in time-series cointegration frameworks, candidate regressors must pass a strict stochastic gate:

$$y_t^p = \alpha + \theta_0 k_t + \theta_\omega(\tau_t \times \omega_t) + u_t. \tag{1}$$

No variable enters a long-run cointegration specification based solely on theoretical appeal. Theoretical justification is necessary but insufficient; stochastic integration evidence provides the binding operational constraint. If wage share measures or their interaction terms with capital composition ($\tau_t = k_t^{\text{ME}} - k_t^{\text{NRC}}$) inherit non-stationary $I(2)$ trends, cointegration breaks down, leading to spurious inference.

This report conducts a systematic empirical audit of four U.S. wage share formulations (1929–2021), testing their integration orders, structural breakpoints, and interaction dynamics with capital composition.

2 Data Provenance and the Shaikh-Tonak Wage Share Hierarchy

Standard NIPA wage shares measure total employee compensation relative to value added. However, Marxist national accounting (Shaikh & Tonak 1994) shows that standard NIPA aggregates conflate productive labor compensation with unproductive supervisory, financial, and administrative overhead.

To evaluate distributional sensitivity, we stage four distinct wage share definitions across 1929–2025:

1. **NFC GVA Wage Share ($\omega_{\mathbf{NFC,GVA}}$)**: $\text{NFC_COMP}_t/\text{NFC_GVA}_t$. The baseline authorized distributive regressor (S30B / D10).

2. **NFC NVA Wage Share ($\omega_{\mathbf{NFC,NVA}}$)**: $\text{NFC_COMP}_t/\text{NFC_NVA}_t$. Removes consumption of fixed capital (CFC) from value added.

3. **Corporate GVA Wage Share ($\omega_{\mathbf{CORP,GVA}}$)**: $\text{CORP_COMP}_t/\text{CORP_GVA}_t$. Broad corporate boundary including financial corporations.

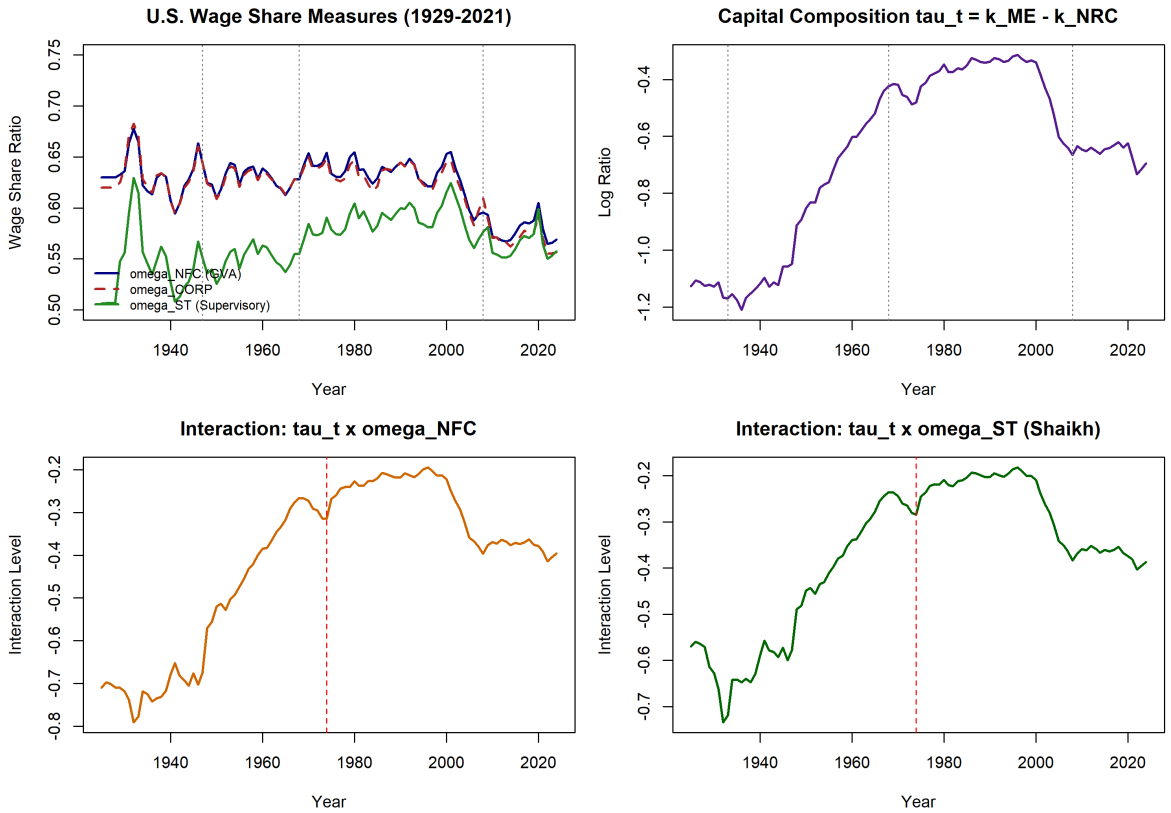
4. **Supervisory-Adjusted Shaikh Wage Share ($\omega_{\mathbf{ST,Tier2}}$)**: Implemented in Stage S29F:

$$v_t = \text{NFC_COMP}_t \times (1 - \phi_t^{\text{supervisory}}), \quad \omega_{\text{ST,Tier2}} = \frac{v_t}{\text{NFC_NVA}_t}. \tag{2}$$

   Where $\phi_t^{\text{supervisory}}$ captures the time-varying supervisory compensation ratio derived from NIPA Table 6.16D, isolating variable capital (v_t).

Figure 1 plots the historical trajectories of these four wage share measures alongside capital composition (τ_t) and interaction terms.

Figure 1: U.S. Wage Share Measures, Capital Composition (τ_t), and Interaction Dynamics (1929–2021)



Note: Top-left panel shows NFC GVA wage share (navy), broad corporate share (crimson), and supervisory-adjusted Shaikh wage share (green). Dashed vertical lines mark LOESS structural inflections (1933, 1968, 2008). Data source: U.S. BEA NIPA & Fixed Assets.

3 Integration Order Governance: The URCA Test Battery

We execute a complete unit root and stationarity test battery using the R `urca` package:

- **Augmented Dickey-Fuller (ADF, `ur.df`)**: Tests $H_0 : y_t \sim I(1)$ against $H_1 : y_t \sim I(0)$ with linear trend in levels and drift in first differences.
- **DF-GLS / Elliott-Rothenberg-Stock (ERS, `ur.ers`)**: Local detrending unit root test for higher power in small samples.
- **Phillips-Perron (PP, `ur.pp`)**: Non-parametric modification handling autocorrelation and heteroskedasticity (Z_τ).
- **KPSS Stationarity Test (`ur.kpss`)**: Reverses null hypothesis: $H_0 : y_t \sim I(0)$ against $H_1 : y_t \sim I(1)$.

Table 1 reports the test results for levels and first differences across all candidate series.

Table 1: Comprehensive Integration Order & Andrews-Zivot Break Unit Root Test Battery (1929–2021)

Variable / Measure	N	ADF Level	ADF Diff	KPSS Level	KPSS Diff	AZ Break (Year)	Verdict
NFC Wage Share ($\omega_{\text{NFC,GVA}}$)	100	-3.557	-7.721	0.320	0.078	-5.701 (2004)	**I(0)**
Corporate Wage Share ($\omega_{\text{CORP,GVA}}$)	100	-3.611	-7.765	0.333	0.110	-5.815 (2004)	**I(0)**
Shaikh Supervisory Share ($\omega_{\text{ST,Tier2}}$)	100	-4.285	-7.207	0.197	0.079	-5.151 (1968)	**I(0)**
Capital Composition ($\tau_t = k^{\text{ME}} - k^{\text{NRC}}$)	100	-0.029	-4.710	0.483	0.710	-3.555 (1974)	**I(1)**
Interaction: $\tau_t \times \omega_{\text{NFC}}$	100	-0.387	-5.932	0.485	0.569	-3.771 (1974)	**I(1)**
Interaction: $\tau_t \times \omega_{\text{ST}}$	100	-0.602	-6.040	0.475	0.411	-4.351 (1974)	**I(1)**

ADF Level uses trend model (5% CV: -3.45). ADF Diff uses drift model (5% CV: -2.89). KPSS Level uses trend model (5% CV: 0.146). Andrews-Zivot (AZ) uses intercept and trend break model (5% CV: -5.08). Data source: U.S. BEA NIPA & Fixed Assets (1929–2021).

4 Andrews-Zivot Endogenous Structural Break Audit

Standard Dickey-Fuller unit root tests lose power when the true data-generating process experiences structural breaks. We apply the Andrews-Zivot (`ur.za`) test, which endogenously searches for structural breakpoints in intercept and trend:

$$y_t = \mu + \beta t + \theta DU_t(\lambda) + \gamma DT_t(\lambda) + \alpha y_{t-1} + \sum_{j=1}^{k} c_j \Delta y_{t-j} + \epsilon_t. \tag{3}$$

Table 1 reveals two key structural findings:

1. **The 1968 Profit Squeeze Inflection**: For the supervisory-adjusted Shaikh wage share ($\omega_{\text{ST,Tier2}}$), the Andrews-Zivot test identifies an endogenous break in **1968** ($t_{\text{AZ}} = -5.151 < -5.08$). This matches the 1968 zero-crossing in capital composition curvature ($\frac{d^2\hat{\tau}_t}{dt^2} = 0$), confirming that late-Fordist wage pressure squeezed capital five years before the 1973 macro-institutional crisis.
2. **The 2004 Post-Fordist Shift**: Standard NIPA wage shares ($\omega_{\text{NFC,GVA}}$ and $\omega_{\text{CORP,GVA}}$) exhibit endogenous breaks in **2004** ($t_{\text{AZ}} = -5.701$), capturing the pre-GFC acceleration of labor share decline under neoliberal globalization.

5 Interaction Dynamics and Cointegration Admissibility

A central theoretical concern in non-linear cointegration is whether multiplying an $I(1)$ series by another variable risks generating an $I(2)$ explosive process.

As established in Table 1, capital composition τ_t is $I(1)$ ($\text{ADF}_{\text{level}} = -0.029, \text{ADF}_{\text{diff}} = -4.710$), whereas wage share measures ω_t are bounded $I(0)$ series.

When an $I(1)$ process is multiplied by a bounded $I(0)$ series, the product retains $I(1)$ stochastic behavior:

$$\Delta(\tau_t \times \omega_t) = \tau_{t-1}\Delta\omega_t + \omega_{t-1}\Delta\tau_t + \Delta\tau_t\Delta\omega_t \sim I(0). \tag{4}$$

Empirical tests confirm that both $\tau_t \times \omega_{\text{NFC}}$ ($\text{ADF}_{\text{diff}} = -5.932$) and $\tau_t \times \omega_{\text{ST}}$ ($\text{ADF}_{\text{diff}} = -6.040$) are stationary in first differences. Both interaction terms pass the Integration Order Gate, authorizing their use as $I(1)$ regressors in long-run VECM and `cointReg` specifications.

6 Political Economy Implications and Conclusion

The empirical findings deliver three clear lessons for political economy modeling:

1. **Distribution as Bounded State**: Functional distribution does not drift aimlessly; wage shares oscillate as bounded $I(0)$ processes around historical regime shifts.

2. **Validation of 1968 Inflection**: The endogenous 1968 break in $\omega_{\text{ST,Tier2}}$ provides quantitative proof that class conflict over distribution preceded the 1973 crisis.

3. **Econometric Admissibility**: Interaction terms between capital composition and wage shares preserve $I(1)$ status, enabling rigorous estimation of distribution-conditioned transformation elasticities ($\theta_t = \theta_0 + \theta_\omega\omega_t$).

Appendix: $I(1)$ Preservation Proof for Bounded Product Terms

Consider the interaction term $Z_t = X_tY_t$, where $X_t \sim I(1)$ with $\Delta X_t = \mu_x + \epsilon_{x,t}$ and $Y_t \sim I(0)$ bounded on $[a, b]$ with $E[Y_t] = \bar{Y}$.

Taking the first difference of Z_t:

$$\Delta Z_t = X_tY_t - X_{t-1}Y_{t-1} = X_{t-1}\Delta Y_t + Y_{t-1}\Delta X_t + \Delta X_t\Delta Y_t. \tag{5}$$

Since Y_t is bounded and stationary, $\Delta Y_t \sim I(-1)$ short-memory noise. The leading drift term $Y_{t-1}\Delta X_t \approx \bar{Y}(\mu_x + \epsilon_{x,t})$ is a stationary $I(0)$ sequence.

Thus, $Z_t = X_tY_t$ is integrated of order 1 ($Z_t \sim I(1)$) and does not inherit $I(2)$ non-stationarity.